

SMART INDIA HACKATHON 2025



TITLE PAGE

- Problem Statement ID – 25004
- Problem Statement Title- Image-based Breed Recognition for Cattle and Buffaloes of India
- Theme- Agriculture, FoodTech & Rural Development
- PS Category- Software
- Team ID- SIH
- Team Name- (InVisionX)



IMAGE BASED BREED RECOGNITION SYSTEM FOR CATTLE AND BUFFALOES IN INDIA

❖ Proposed Solution (Idea/ Solution/ Prototype) :-

Present **Label Blend**, a **synthetic data generation tool** that **automatically creates large volumes of labeled images** (Bounding Box / Segmentation-based datasets) of **cattle and buffalo** under varying **poses, environments, and lighting conditions**. This **augments and enriches datasets** for **robust breed classification**.

How It Addresses the Problem:

- ✓ **Overcomes limited and inconsistent training data**
- ✓ **Ensures breed recognition accuracy under real-world field conditions**
- ✓ **Integration with Bharat Pashudhan App (BPA): feeding high-quality synthetic datasets into model training pipelines**
- ✓ **Improves AI model generalization across diverse cattle and buffalo breeds**

Innovation & Uniqueness:

- **Customizable environment and pose generation** → enhances dataset diversity
- **Auto-labeled synthetic data** → accelerates AI/ML development lifecycle
- **Scalable solution** → adaptable across **indigenous and crossbred variations**
- Strengthens **Computer Vision, Deep Learning, and Image Recognition** performance in livestock management



TECHNICAL APPROACH

Implementation Roadmap -

Technologies we Use in this Project:

Python for (Backend & AI/ML, Data Processing, Scripting), HTML5, Tailwind CSS, JavaScript (Frontend & Web Development)

Frameworks:

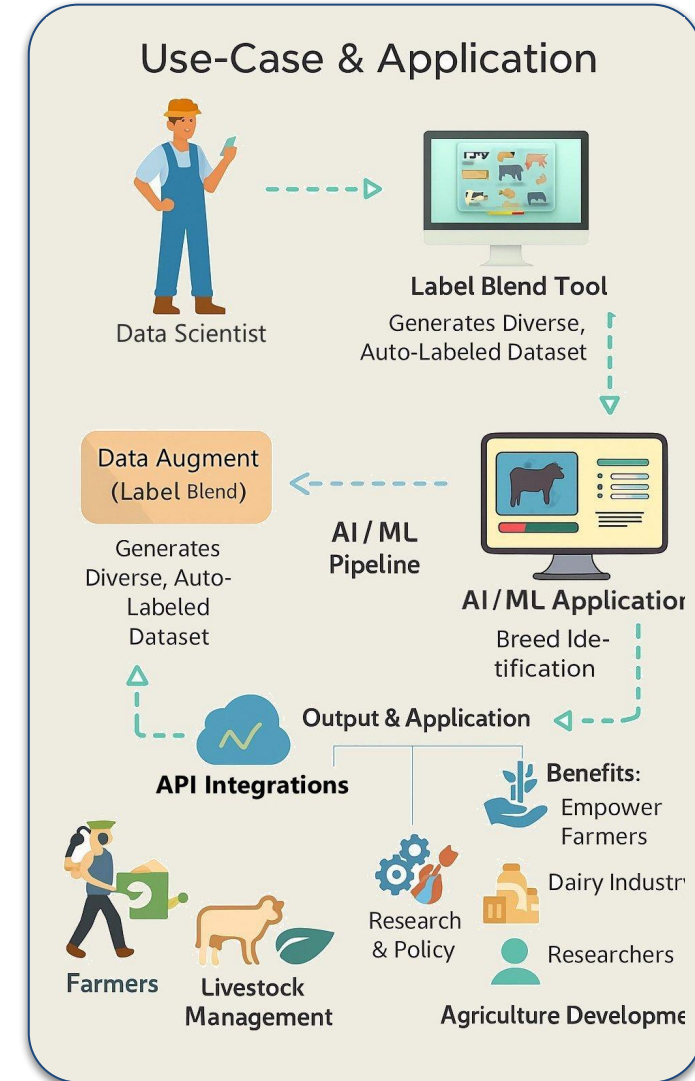
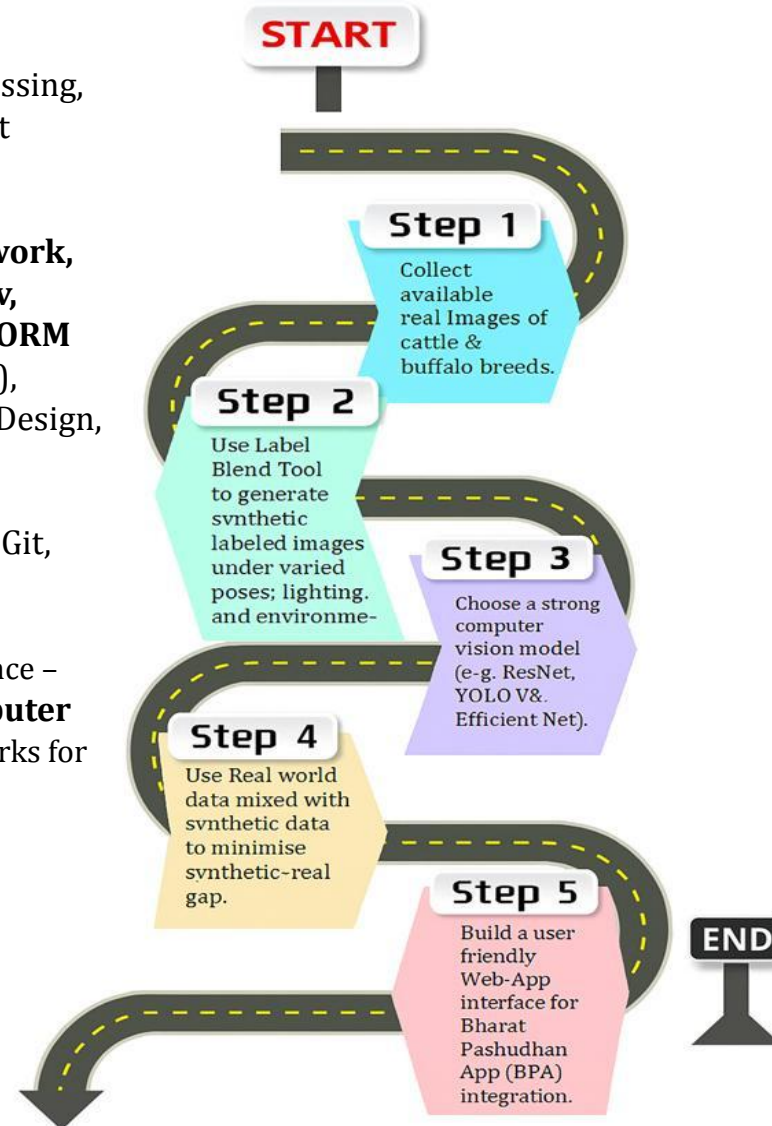
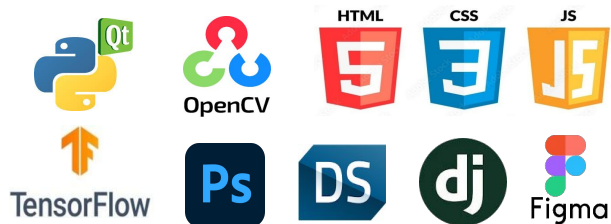
Django (Full Stack), React, Qt Framework, TensorFlow, Keras, REST APIs OpenCv, Spring Framework, MySQL, SQLite & ORM PySide (Open Source GUI Development), Qt Framework (Cross-platform UI/UX Design, GUI Applications)

Tools:

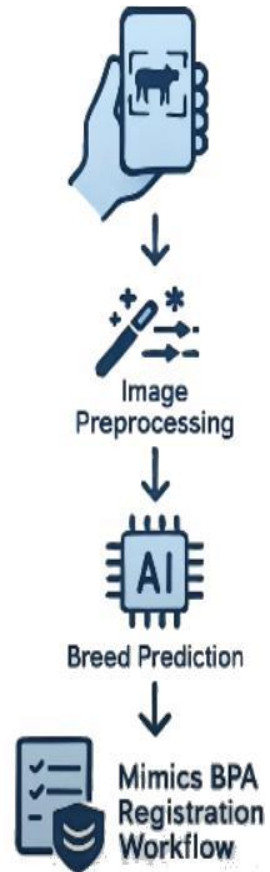
VS Code, Jupyter, Collab, GitHub, Figma, Git, Insomnia for API, Qt Design Studio

Inference Models:

Opencv + YOLOv8-Seg (You Only Look Once – Object Detection & Segmentation) + Computer Vision (ResNet-50 (Deep Residual Networks for Image Classification))



API for BPA (App)

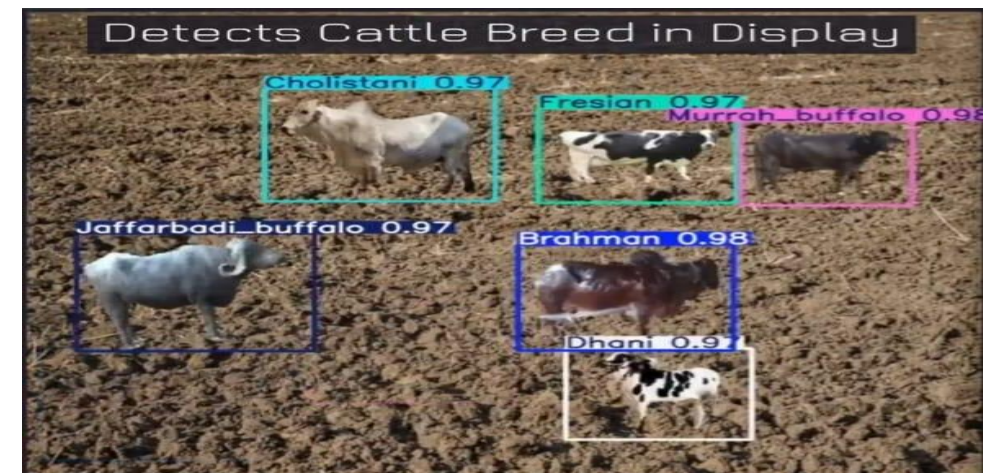


Feasibility:

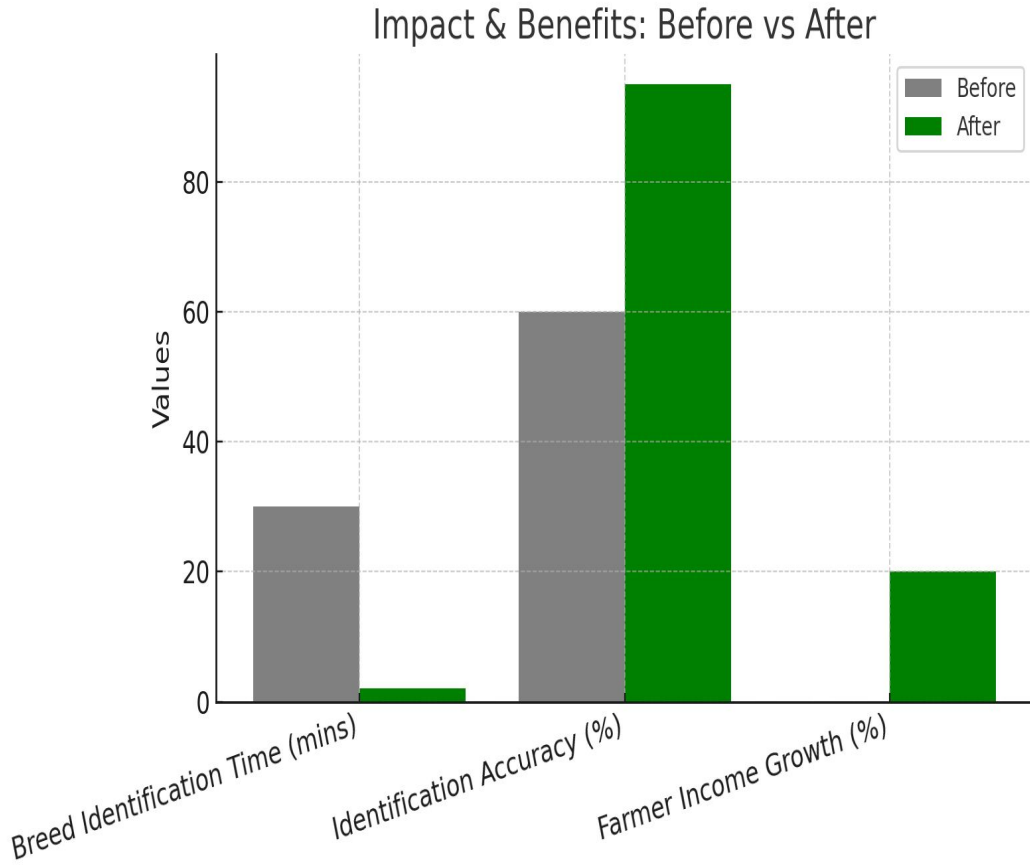
- **Team Expertise:** Python, Django, REST APIs, SQL, Qt Framework, Open Source development, UI/UX, Data Science, Deep Learning.
- **Proof of Concept:** Success of existing **object detection & image recognition systems** validates feasibility.

Challenges	Solutions
Data Diversity & Quality Issues – Breed images vary in lighting, pose, environment → impacts model accuracy.	Deploy Label Blend Studio for synthetic dataset generation, data augmentation, and auto-labeling to ensure high-quality, diverse training data.
Adoption Barriers for Field Workers (FLWs) – Limited technical literacy & low connectivity in rural areas.	Build offline-first mobile app with intuitive UI/UX, guided image capture, and automated feedback for easy adoption
Model Accuracy at Scale – Risk of misclassification in crossbred variations .	Implement continuous model retraining with real + synthetic datasets, active learning pipelines, and computer vision model optimization

Project Screenshots (Main):



Stakeholder	Impact	Benefits
Farmers	Accurate breed recognition at the field level	Fair pricing, better health monitoring, improved productivity
Government & Policy Makers	Reliable and standardized livestock data	Stronger BPA, accurate census, efficient subsidy & scheme implementation
Veterinarians & Researchers	Access to diverse real + synthetic datasets	Supports disease prediction, breed improvement, genetic research
Economy & Rural Development	Scalable solution for millions of cattle	Strengthens supply chains, boosts rural livelihoods, contributes to GDP
Overall Ecosystem	Data-driven livestock management	Transparency, efficiency, smarter rural development



- **FAO – Livestock Biodiversity Report**

<http://www.fao.org/livestock-biodiversity/en/>

- **BPA (Bharat Pashudhan Portal)**

<https://bharatpashudhan.ndlm.co.in/>

- **ICAR – Cattle & Buffalo Breeds in India**

<https://icar.org.in/content/cattle-and-buffalo-breeds>

- **Government of India – Livestock Census (DAHD)**

<http://dahd.nic.in/division-type/statistics-livestock-census>

- **NDDDB – National Dairy Development Board**

<https://www.nddb.coop/services/animalbreeding>

- **Springer – Artificial Intelligence in Agriculture: Cattle Breed Recognition**

<https://link.springer.com/journal/43681>

- **Elsevier – Deep Learning for Livestock Identification (Computers & Electronics in Agriculture)**

<https://doi.org/10.1016/j.compag.2020.105514>

- **Case Study – Dairy Co-ops**

 Field case studies on misclassification impact (milk yield pricing, income loss).

Prototype Video Link - <https://youtu.be/Pplf59qdlj4>



सत्यमेव जयते

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